

Usama Mehmood

Department of Computer Science
Information Technology University
ASTP, Ferozpur Rd., Lahore

Email: usamamehmood@itu.edu.pk
Phone: 0323-4853535
[LinkedIn](#) | [Google Scholar](#)

RESEARCH AREAS

Safety in Cyber-Physical Systems, Autonomous Vehicles and Multi-Agent Robotics, Urban Mobility and Traffic Simulation

EDUCATION

09/2015 to 01/2021: **Stony brook University**, NY, US Ph.D., Computer Science
Dissertation: Advances in Safety Assurances
for Multi-Agent Systems
Advisor: Prof. Scott A. Smolka
09/2015 to 09/2019: **Stony brook University**, NY, US M.S., Computer Science
09/2010 to 06/2014: **LUMS**, Lahore, PAK B.S, Electrical Engineering

WORK EXPERIENCE

10/2021 to present: **Information Technology University**, Computer Science Dept.
Assistant Professor
06/2021 to 04/2024: **Educative Inc.**
Consultant for Online Course Design and Development (part-time)
01/2016 to 01/2021: **Stony Brook University**, Computer Science Dept.
Graduate Research Assistant with Professor Scott A. Smolka
05/2019 to 08/2019: **Facebook Inc.**, New York City, NY, US
Software Engineer Intern with Instagram Machine Learning Team
05/2018 to 08/2018: **The Mathworks**, Novi, MI, US
Software Engineer Intern with Simulink Automotive Team
01/2015 to 08/2015: **LUMS**, Computer Vision Lab
Research Assistant with Dr. Sohaib Khan

COURSES TAUGHT

All courses were taught at Information Technology University of the Punjab (ITU).

1. Analysis of Algorithms, Fall 2021, Fall 2022, Fall 2023, Fall 2024
2. Computational Geometry, Spring 2022
3. Cyber-Physical Systems, Spring 2023

4. Reinforcement Learning, Fall 2025
5. Theory of Automata, Spring 2024, Spring 2025

PUBLICATIONS

Ph.D. Thesis

1. Usama Mehmood. *Advances in Safety Assurances for Multi-Agent Systems*. Stony Brook University, Stony Brook, NY, January 2021.

Journal Publications

1. Hafiz Rizwan Iqbal, Muhammad Sharjeel, Jawad Shafi, Usama Mehmood, Saeed Ul Hassan, Agha Ali Raza. Urdu Paraphrased Text Reuse and Plagiarism Detection using Pre-trained Large Language Models and Deep Hybrid Neural Networks. *Multimedia Tools and Applications*, April 2025.
2. Hafiz Rizwan Iqbal, Muhammad Sharjeel, Jawad Shafi, Usama Mehmood, Agha Ali Raza. Urdu Sentential Paraphrased Plagiarism Detection Using Large Language Models. *Transactions on Asian and Low-Resource Language Information Processing*, July 2025.
3. Usama Mehmood, Sanaz Sheikhi, Stanley Bak, Scott A. Smolka, and Scott D. Stoller. The Black-Box Simplex Architecture for Runtime Assurance of Multi-Agent CPS. *Innovations in Systems & Software Engineering Journal*, a NASA journal, March 2024.
4. Usama Mehmood, Shouvik Roy, Amol Damare, Radu Grosu, Scott A. Smolka, and Scott D. Stoller. A Distributed Simplex Architecture for Multi-Agent Systems. *Journal of Systems Architecture* 134, January 2023.

Conference and Workshop Publications

1. Usama Mehmood, Sanaz Sheikhi, Stanley Bak, Scott A. Smolka, and Scott D. Stoller. The Black-Box Simplex Architecture for Runtime Assurance of Autonomous CPS. In *Proceedings of the 14th NASA Formal Methods Symposium (NFM 2022)*, volume 13260 of Lecture Notes in Computer Science, pages 231-250. Springer-Verlag, 2022.
2. Usama Mehmood, Scott D. Stoller, Radu Grosu, Shouvik Roy, Amol Damare, and Scott A. Smolka. A Distributed Simplex Architecture for Multi-Agent Systems. In *Proceedings of the 7th International Symposium on Dependable Software Engineering: Theories, Tools and Applications (SETTA 2021)*, volume 13071 of Lecture Notes in Computer Science, pages 239-257. Springer-Verlag, 2021.
3. Usama Mehmood, Stanley Bak, Scott A. Smolka and Scott D. Stoller. Safe CPS from Unsafe Controllers (Extended Abstract). In *Proceedings of the 2021 Workshop on Computation-Aware Algorithmic Design for Cyber-Physical Systems (CAADCPS)*. ACM Press, 2021. This is a preliminary version of the work described in our NFM 2022 paper.
4. Shouvik Roy, Usama Mehmood, Radu Grosu, Scott A. Smolka, Scott D. Stoller, and Ashish Tiwari. Distributed Control for Flocking Maneuvers via Acceleration-Weighted Neighborhooding. In *Proceedings of the 2021 American Control Conference (ACC 2021)*, pages 2745-2750. IEEE Press, 2021.

5. Usama Mehmood, Scott D. Stoller, Radu Grosu, and Scott A. Smolka. Collision-Free 3D Flocking Using the Distributed Simplex Architecture. In *Formal Methods for Outer Space: Essays Dedicated to Klaus Havelund on the Occasion of His 65th Birthday*, volume 13065 of Lecture Notes in Computer Science. Springer-Verlag, 2021.
6. Usama Mehmood, Shouvik Roy, Radu Grosu, Scott A. Smolka, Scott D. Stoller, and Ashish Tiwari. Neural Flocking: MPC-based Supervised Learning of Flocking Controllers. In *Proceedings of the 23rd International Conference on Foundations of Software Science and Computation Structures (FoSSaCS 2020)*, volume 12077 of Lecture Notes in Computer Science. Springer-Verlag, 2020.
7. Shouvik Roy, Usama Mehmood, Radu Grosu, Scott A. Smolka, Scott D. Stoller, and Ashish Tiwari. Learning Distributed Controllers for V-Formation. In *Proceedings of the 1st IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS 2020)*. IEEE, 2020.
8. Usama Mehmood, Nicola Paoletti, Dung Phan, Radu Grosu, Shan Lin, Scott D. Stoller, Ashish Tiwari, Junxing Yang, and Scott A. Smolka. Declarative vs Rule-based Control for Flocking Dynamics. In *Proceedings of the 2018 ACM Symposium on Applied Computing (SAC 2018)*, Intelligent Robotics and Multi-Agent Systems (IRMAS) track. ACM Press, 2018.

OPEN SOURCE CONTRIBUTION

A RanSaC algorithm for fitting cuboids on 3D point-clouds. [[GIT](#)] [[MATLAB](#)]

PRESENTATIONS

1. The Black-Box Simplex Architecture for Runtime Assurance of Autonomous CPS, NASA Formal Methods Symposium, Caltech, Pasadena, US, May 25, 2022.
2. A Distributed Simplex Architecture for Multi-agent Systems, Symposium on Dependable Software Engineering (SETTA 2021), Beijing, China, November 26, 2021.
3. Collision-Free 3D Flocking Using the Distributed Simplex Architecture, 9th International Symposium on Leveraging Application of Formal Methods, Verification, and Validation (ISOLA 2021), Rhodes, Greece, October 24, 2021.
4. Declarative vs Rule-based Control for Flocking Dynamics, 33rd ACM/SIGAPP Symposium On Applied Computing, Pau, France, April 11, 2018.

Invited Talks at Universities and Research Institutes

1. Runtime Assurance for Multi-Agent Robotic Systems, Information Technology University (ITU), Lahore, August 21, 2021
2. Advances in Safety Assurances for Multi-Agent Systems, Department of Electrical Engineering, Lahore University of Management Sciences (LUMS), Lahore, June 30, 2021
3. Runtime Assurance for Cyber-Physical Systems, Department of Computer Science, Lahore University of Management Sciences (LUMS), Lahore, December 21, 2020

AWARDS AND HONOURS

Graduate Student Fellowship Award at Stony Brook University (\$4000)

SERVICE TO THE UNIVERSITY & DEPARTMENT

Undergraduate CS Program Accreditation Planning for NCEAC Visit, 2025

Grievance Redressal Committee, 2023-present

PhD students Interview Panel, 2021-present

Teaching Fellows Interview Panel, 2021-present

Admission Test Preparation 2021-present

Self-Assessment Team for graduate CS programs, 2022-present

STUDENT SUPERVISION

- Supervised 12 Bachelor's Final Year Projects
- Supervised 3 Master's thesis projects
- Evaluated 20 MS thesis projects
- Serves on Graduate Examination Committee of four PhD students